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SARDAR PATEL UNIVERSITY

M.Sc. CA&IT Examination - Nov 2019

1st Semester NC (CBCS) (Regular)

Saturday , Date: 23-11-2019

Session : Morning

Time : 10:00 AM to 01:00 PM

Course Code: PS01CHT34

Course Title : Advanced Computer Networking

Total Marks: 70

Q.1 Multiple Choice Questions. (Attempt All)

[08]

1. Which of the following are classes of routing algorithms?
A. Adaptive C. Session
B. Non-Adaptive D. Both A and B
2. Which of the following is responsibility of router?
A. Encrypt packets and send them back to sender C. Find optimal route to send packets
B. Decrypt the packets and send them back to sender D. Detect reason for hardware failure of connected nodes
3. In shortest path routing algorithm, the nodes are labelled _____ initially.
A. Infinite C. One
B. Negative One D. Zero
4. Leaky bucket technique is used for _____.
A. Flow control C. Error control
B. Congestion control D. Optimal path
5. Which of the following is the part of URL?
A. Protocol C. DNS Name
B. Local Name of webpage D. All of the above
6. SMTP protocol is used to _____.
A. encrypt file C. download webpage
B. send e-mail D. receive command
7. Out of 19 stages _____ number of stages in DES algorithm are functionally identical but uses unique key.
A. 16 C. 12
B. 07 D. 03
8. In s-box device, s stands for _____.
A. subtraction C. simulation
B. substitution D. summation

Q.2 Short Questions. (Attempt any seven)

[14]

1. What is Session routing?
2. What is Selective flooding technique?
3. What is Internetworking?
4. State advantages of Concatenated Virtual Circuits over datagram approach.
5. State various need for Fragmentation.
6. What is Resolver?
7. What is User agent and Message transfer agent?
8. What is Plaintext and Ciphertext?
9. What is difference between Passive and Active intruder?

Q.3[a] Explain Shortest Path Routing algorithm in detail with example. [6]

Q.3[b] Explain Distance Vector Routing algorithm along with Count to Infinity problem. [6]

OR

Q.3[b] What is Congestion? Explain in detail congestion prevention policies. [6]

Q.4[a] Explain issues arising when different types of networks are connected. [6]

Q.4[b] Explain different Fragmentation methods in detail. [6]

OR

Q.4[b] Explain the concept of Tunneling using labeled diagram. [6]

Q.5[a] Explain functions of e-mail in detail. [6]

Q.5[b] Explain step involved in server side processing when user queries for particular webpage. [6]

OR

Q.5[b] What are Generic and Country domains? Explain components of DNS and information stored in resource record format of DNS. [6]

Q.6[a] Explain DES Algorithm in detail. [6]

Q.6[b] Explain Substitution and Transposition cipher in detail. [6]

OR

Q.6[b] Write steps for RSA algorithm. Find cipher text for plain text "S", for given values of $p=3$, $q=11$ and $e=3$. [6]

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